Hall Ticket Number:

MR-21

Code: B414CS4S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Java Programming

(Common to CSE and IT)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Define object. Write the syntax for creating an object.	2	1	1	1
1. b)	Differentiate method overloading and method overriding.	2	2	1	2
1. c)	Mention any four inbuilt packages in java.	2	1	2	1
1. d)	Give the syntax of try and catch blocks in exception handling.	2	1	2	1
1. e)	Differentiate thread based multitasking and process based multi-tasking.	2	2	3	2
1. f)	Explain the usage of Console class in java with an example.	2	2	3	1
1. g)	List any four inbuilt methods of Stack class.	2	1	4	1
1. h)	Explain the usage of Scanner class with an example.	2	2	4	1
1. i)	Compare and contrast applets and applications.	2	2	5	2
1. j)	State any two limitations of AWT.	2	1	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain about java buzzwords.	5	1	1	1
2. b)	Write a java program that illustrates constructor overloading.	5	3	1	1
	OR				
2. c)	Discuss about the usage of final keyword with inheritance.	5	2	1	1
2. d)	Differentiate class and an abstract class using suitable examples.	5	3	1	2
3. a)	Define package. Explain how packages can be created and imported with suitable examples.	5	2	2	1
3. b)	Explain how multiple inheritance is achieved through interfaces in java using suitable examples.	5	4	2	2
	OR				
3. c)	Discuss about the usage of finally keyword in exception handling with examples.	5	2	2	1
3. d)	Write a java program that illustrates the creation of own exception subclasses.	5	3	2	2
4. a)	Explain the creation of thread by extending Thread class using suitable examples.	5	2	3	1
4. b)	Discuss about inter thread communication in java.	5	3	3	2
	OR				
4. c)	Describe the usage of generics in java.	5	1	3	1
4. d)	Write a java program that copies the content of one file to another file.	5	3	3	2
5. a)	Explain the usage of ArrayList class using suitable java program.	5	2	4	1
5. b)	Write a java program that illustrates the usage of StringTokenizer class.	5	3	4	1
	OR				

P.T.O.

5. c)	Discuss about for each alternative with examples.	5	2	4	2
5. d)	Explain how to access a Collection via an interface using suitable java program.	5	3	4	1
6. a)	Explain about MVC architecture in detail.	5	1	5	1
6. b)	Discuss about different layout managers in java.	5	2	5	2
	OR				
6. c)	Write a java program that illustrates passing parameters through applets.	5	3	5	1
6. d)	Explain Adapter classes in java with suitable examples.	5	3	5	1

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome

Hall Ticket Number:

MR-21

Code: B414CB4S3

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Supplementary Examinations, December 2024/January 2025

Object Oriented Programming using Java

(Common to CSB, CSM and CSD)

Time: 3 Hours | ***Answer ALL Questions*** | Max. Marks: 70

Part-A (10 x 2 = 20 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	List the core OOP's concepts.	2	1	1	1
1. b)	What is Constructor overloading?	2	2	1	1
1. c)	What is Exception handling?	2	1	2	1
1. d)	Define Package. How to create Package?	2	2	2	1
1. e)	Compare thread based and process based multi-tasking.	2	2	3	2
1. f)	Define Enumerations with example.	2	2	3	1
1. g)	What is Collection?	2	1	4	1
1. h)	Describe StringTokenizer.	2	2	4	1
1. i)	Define an Applet.	2	1	5	1
1. j)	What is Layout Manager? List Layout Managers.	2	2	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Describe in detail about the control flow statements in Java with suitable examples.	5	2	1	1
2. b)	Describe the following in detail (i)Data Abstraction and encapsulation. (ii)Polymorphism and dynamic binding	5	2	1	1
	OR				
2. c)	Discuss the following: (i) using super keyword. (ii) Data types in Java	5	2	1	1
2. d)	Analyse and Develop a simple Java program to sort the given numbers in increasing order.	5	3	1	3
3. a)	Illustrate what is meant by package? How it is created and implemented in Java.	5	2	2	2
3. b)	What is an interface? Explain how to define and implement by taking suitable Example.	5	2	2	2
	OR				
3. c)	Summarize what is finally class. How to catch exception? Write an example.	5	2	2	2
3. d)	Discuss in detail about exception handling constructs and write a program to illustrate Divide by zero exception.	5	2	2	3
4. a)	Describe the lifecycle of thread in Java with an example.	5	2	3	1
4. b)	Explain the concept of readers writers' problem and implement the concept using the suitable class and methods available in Java.	5	3	3	2
	OR				
4. c)	Explain the following (i)thread priorities. (ii)Explain how threads are created in Java.	5	3	3	2
4. d)	(i)Illustrate the concept of synchronization in thread. (ii) The Stream Classes-Byte streams and Character streams.	5	3	3	1

P.T.O.

5. a)	Explain in detail about collection interfaces and Class hierarchy.	5	3	4	1
5. b)	Implement a Java program which shows the HashSet and Iterator.	5	3	4	2
	OR				
5. c)	Explain about Dictionary properties and Hash table.	5	2	4	1
5. d)	Explain More Utility classes i) Random ii) Scanner.	5	4	4	1
6. a)	Write a Java Program for Handling Mouse events.	5	4	5	2
6. b)	Explain about Different Swing Components.	5	3	5	1
	OR				
6. c)	Explain on AWT Event Hierarchy.	5	2	5	1
6. d)	Examine the following in detail (i)Handling a TextField. (ii)Using a TexArea.	5	3	5	2

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome